

Imprecision Attenuates Updating*

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Abstract

This paper studies how imprecision in noisy signals attenuates Bayesian posterior means toward the prior mean, a property that underlies many comparative-statics results in information economics. We show this effect holds for symmetric location experiments, using a newly introduced precision order. Our main result is that this order is characterized by the attenuation effect: the posterior mean under noise density $\tilde{g}$ is closer to the signal than under noise density g for all signal realizations and all symmetric, log-concave priors, if and only if $\tilde{g}$ is more precise than g . We apply our result to derive comparative statics for prior precision, the value of information, and overconfidence.

Keywords: signal extraction, reversion to the mean, underreaction, central tendency effect, anchoring and adjustment, cognitive imprecision

JEL Classifications: C11, D83, D84

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1 Introduction

In many economic contexts, agents fail to respond optimally to fundamental variables: firms and households adjust only partially to macroeconomic changes (Sims 2003), consumers underreact to non-salient taxes (Chetty, Looney, and Kroft 2009), and numerical estimates are biased toward default values (Tversky and Kahneman 1974). Beyond economics, psychophysics documents a central-tendency effect: perceptual judgments are biased toward the mean of the stimulus distribution (Hollingworth 1910; Stevens and Greenbaum 1966).

Such attenuated behavior is often explained through imprecise information. In one version of this explanation, agents base their choices on noisy signals of the fundamentals, where noise arises from observational errors. In a second version, agents observe the fundamentals but the cognitive process that translates fundamentals to actions is noisy, reflecting cognitive constraints. The latter explanation has gained attention in the growing literature on *cognitive imprecision* (Gabaix 2019; Woodford 2020; Enke and Graeber 2023; Enke, Graeber, Oprea, and Yang 2026), which shows how noisy information processing can account for a wide range of economic phenomena.

These explanations share a core mechanism: noise attenuates Bayesian updating, which compresses behavior toward a default action. This mechanism is usually formalized using the *normal-normal model*, where the agent observes a noisy signal $S = X + \varepsilon$. Here, the state $X \in \mathbb{R}$ is normally distributed, and $\varepsilon \in \mathbb{R}$ is independent, normal noise. The agent’s posterior mean is then compressed toward the prior mean, and more so when the signal is less precise: the posterior mean lies between the signal and the prior mean, and *the posterior mean is closer to the prior mean the larger the variance of ε* .¹ We refer to this effect as *imprecision attenuates updating*. If behavior is based on the posterior mean, then such attenuation of updating leads to attenuation of behavior. In fact, a key piece of evidence for cognitive imprecision relies on this effect: subjects reporting higher cognitive uncertainty exhibit more attenuated behavioral responses (Enke and Graeber 2023; Enke, Graeber, Oprea, and Yang 2026).

Despite its empirical relevance, it remained unknown whether the attenuation effect extends beyond normal distributions, which rely on strong parametric assumptions. While normality can be justified in specific contexts—e.g., via the central limit theorem or rational inattention with a normal prior and quadratic loss—these cases are limited. A non-parametric class of signal structures that preserves the attenuation effect would offer a more robust foundation for interpreting empirical evidence as a consequence of imprecise information.

To address this gap, we prove the imprecision-attenuates-updating effect holds in all additive-noise models (location experiments) with symmetric noise under a symmetric, log-concave prior: for any signal realization, the posterior mean moves toward the prior mean as the signal becomes less precise (Theorem 1).

¹Formally, if $X \sim \mathcal{N}(\mu, \sigma_X^2)$ and $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, then

$$\mathbb{E}[X|S = s] = \lambda s + (1 - \lambda) \mu, \quad \lambda = \frac{\sigma_X^2}{\sigma_X^2 + \sigma_\varepsilon^2}.$$

For any signal realization s , the larger σ_ε^2 , the closer the posterior mean is to the prior mean.

To formalize this result, we introduce a notion of precision, which we call the *precision order*. This order requires the noise distribution of a more precise signal to be farther from zero in the sense of likelihood-ratio domination: noise density $\tilde{g}$ is more precise than noise density g if the likelihood ratio $\tilde{g}(x)/g(x)$ decreases as x moves away from 0 (see Figure 1). Concretely, we show that scaling up a log-concave noise term produces a less precise signal under this definition. As a result, the precision order applies to log-concave location-*scale* experiments.²

We show that the precision order is not only sufficient but also *necessary* for the imprecision-attenuates-updating effect. The precision order is thus characterized by the effect. If a noise density $\tilde{g}$ is *not* more precise than another noise density g , then for some symmetric, log-concave prior f and some signal realization, $\tilde{g}$ induces *more* attenuation than g . We further show that neither Blackwell nor Lehmann dominance implies the precision order, so even a signal that is dominates another in these orders can induce more attenuation.

Beyond extending the attenuation effect to a broad class of distributions, the paper pins down which distributional conditions are essential, symmetry foremost among them. The necessity direction of Theorem 1 shows that the noise densities must be symmetric for the attenuation ordering to hold across all symmetric, log-concave priors. Symmetry is equally essential for the more basic property that the posterior mean lies between the prior mean and the signal. Chambers and Healy (2012) show that without symmetric prior and noise density, the posterior mean can fall outside this interval, so that agents overreact or update in the opposite direction.

The normal-normal model is pervasive in economics, and comparative statics in signal precision are among its most common uses. Because our result frees these comparative statics from the normality assumption, it has implications well beyond behavioral attenuation. We establish comparative statics for the median and the average posterior mean conditional on the state. We also study how prior precision, rather than signal precision, shapes updating. A more precise prior increases attenuation and lowers the value of information, the opposite of a more precise signal.

Related Literature This paper introduces a new information order and provides comparative-statics results for signal structures with additive independent noise, which are called location experiments.

Prior work has explored orderings on location experiments related to the value of information. Boll (1955) shows that one location experiment Blackwell-dominates another if and only if the latter's noise term can be derived by adding an independent noise term to the former's. This condition is very restrictive, making location experiments comparable only in limited cases. In response, Lehmann (1988) introduced the less restrictive Lehmann order to rank location experiments for *monotone* decision problems. Our precision order is distinct from the Blackwell order, neither implying nor being implied by it. The precision order strengthens the Lehmann order.

Other comparative statics results have been established for location experiments with log-concave noise distributions, which satisfy the monotone likelihood-ratio property. Milgrom (1981) shows

²Furthermore, we demonstrate that a weaker condition than log-concavity suffices, which holds for many non-log-concave noise distributions (Lemma 1).

that the strict monotone likelihood-ratio property ensures higher signals constitute “good news,” in the sense of producing first-order stochastically dominant posteriors, for any prior. Conversely, Dawid (1973) examines violations of the “good news”-property under heavy-tailed noise distributions, showing that if the noise is sufficiently heavy-tailed relative to the prior, extreme observations are “rejected,” yielding posteriors that revert to the prior. Together with our results, this highlights log-concave location-scale experiments are a well-behaved class of signal structures that retains key properties of the normal-normal model. Kartik, Lee, and Suen (2021) provides a comparative statics result with economic applications for agents with heterogeneous priors, termed “information validates the prior,” in which a more informative experiment brings another agent’s posterior mean closer to one’s prior mean.

2 Setup

We study Blackwell experiments (signal structures), where the signal equals the one-dimensional state plus independent noise, which are called *location experiments*. A Blackwell experiment is a family $(G_x(\cdot))_{x \in \mathbb{R}}$, where $G_x(\cdot)$ is the cumulative distribution function (CDF) over signals $s \in \mathbb{R}$ conditional on state x . A Blackwell experiment where $G_x(s) = G(s - x)$ for all $s, x \in \mathbb{R}$ is a location experiment. We restrict to location experiments where G admits a density g and assume the agent’s prior admits a density. Throughout the paper we maintain the following regularity assumption.

Assumption 1. *All considered densities f are strictly positive, continuously differentiable, and have finite first absolute moment $(\int f(x)|x|dx < \infty)$.*

Throughout the paper, $f, \tilde{f}, g, \tilde{g}$ denote densities. In particular, when we need to distinguish between prior and noise densities, $f, \tilde{f}$ will denote prior densities and $g, \tilde{g}$ will denote noise densities.

The posterior mean upon observing signal s is

$$m(s, f, g) := \frac{\int x f(x) g(s - x) dx}{\int f(x) g(s - x) dx}.$$

Assumption 1 ensures that $m(s, f, g)$ is well-defined for any s .

We will make use of the following assumptions on densities.

Definition 1. *The density f is symmetric around μ if for all $x > 0$, $f(\mu + x) = f(\mu - x)$. The density f is symmetric if there exists $\mu \in \mathbb{R}$ such that f is symmetric around μ .*

Definition 2. *The density f is quasi-concave if for all $x, x' \in \mathbb{R}$ and $\alpha \in (0, 1)$, $f(\alpha x + (1 - \alpha)x') \geq \min\{f(x), f(x')\}$.*

Definition 3. *The density f is log-concave if $\log f$ is concave.*

Given the assumption that densities are positive, log-concavity implies quasi-concavity.

Symmetric and quasi-concave prior and noise densities are particularly relevant for our study. The reason is the following result, which is a strengthening of Chambers and Healy (2012). The

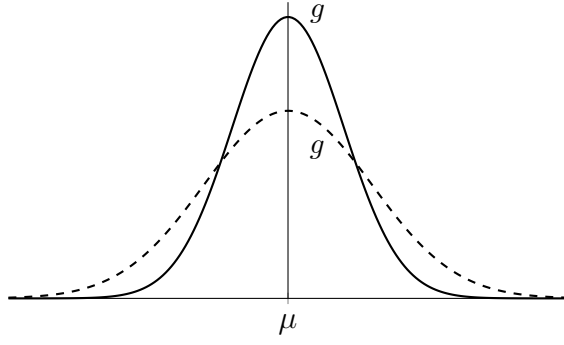


Figure 1: The densities of two distributions $\tilde{g} \sim \mathcal{N}(0, 1)$ and $g \sim \mathcal{N}(0, 1.5^2)$ where $\tilde{g}$ is more precise.

third statement of the proposition is new. All proofs are relegated to the appendix. For any f let $\mu_f := \int x f(x) dx$ denote the induced mean.

Proposition 1 (Chambers and Healy 2012). *If the prior density f is symmetric and quasi-concave and the noise density g is symmetric around 0 and quasi-concave, then*

$$\begin{aligned} \forall s \leq \mu_f: \quad s &\leq m(s, f, g) \leq \mu_f, \\ \forall s \geq \mu_f: \quad s &\geq m(s, f, g) \geq \mu_f. \end{aligned} \tag{1}$$

If (1) holds for all symmetric around 0 and quasi-concave g , then f is symmetric and quasi-concave. If (1) holds for all symmetric and quasi-concave f , then g is symmetric around 0 and quasi-concave.

Symmetry and quasi-concavity of both densities are therefore necessary and sufficient for the posterior mean to lie robustly between the signal s and the prior mean μ_f , robustly meaning for every symmetric, quasi-concave density in the other role.

3 Main Result

A challenge to formalizing the idea that imprecision attenuates updating is to find the right order on signal structures. We show that the following new order, which we call the *precision order*, is characteristic of our desired comparative-statics result. We identify location experiments with their noise densities g . Given Proposition 1, we restrict attention to symmetric noise densities.

Definition 4 (Precision Order). *Let g and $\tilde{g}$ be symmetric around μ . Then $\tilde{g}$ is more precise than g if the likelihood-ratio $\tilde{g}(x)/g(x)$ is nonincreasing in x for $x > \mu$.*

Note that by the symmetry of densities, the definition implies the likelihood ratio $\tilde{g}(x)/g(x)$ is nondecreasing in x for $x < \mu$. For noise densities symmetric around 0, the precision order says $|\tilde{\varepsilon}|$ is smaller than $|\varepsilon|$ in the likelihood-ratio order, where $\varepsilon \sim g$ and $\tilde{\varepsilon} \sim \tilde{g}$. In words, the more precise location experiment has a noise term closer to 0 in the sense of likelihood-ratio domination. Figure 1 gives an example of two noise distributions ranked by the precision order. The more precise distribution has a higher density around zero, which falls faster as we move away from zero. Below

we show that the precision order has the intuitive property that scaling up a log-concave density makes it less precise, and relate the precision order to existing information orders.

Our main result characterizes the precision order through the *imprecision attenuates updating* effect. Recall that μ_f denotes the induced mean for any density f .

Theorem 1. *For noise densities g and $\tilde{g}$, the following are equivalent:*

1. *Both g and $\tilde{g}$ are symmetric around 0, and $\tilde{g}$ is more precise than g .*
2. *For every symmetric and log-concave prior density f :*

$$\begin{aligned} \forall s \leq \mu_f: \quad & s \leq m(s, f, \tilde{g}) \leq m(s, f, g) \\ \forall s \geq \mu_f: \quad & s \geq m(s, f, \tilde{g}) \geq m(s, f, g). \end{aligned} \tag{2}$$

The theorem shows that the precision order is both sufficient and necessary for the attenuation ordering (2). For sufficiency, whenever g and $\tilde{g}$ are symmetric around 0 and $\tilde{g}$ is more precise than g , the posterior mean under $\tilde{g}$ is closer to the signal realization than the posterior mean under g , for every signal realization and every symmetric, log-concave prior. Moreover, both posterior means lie on the same side of the signal s as the prior mean μ_f . For necessity, the precision order cannot be weakened. If (2) holds for every symmetric, log-concave prior, then the noise densities must be symmetric around 0 and $\tilde{g}$ must be more precise than g .

If the noise density is also quasi-concave, Proposition 1 places both posterior means weakly between the signal and the prior mean. The ordering (2) then becomes

$$s \leq m(s, f, \tilde{g}) \leq m(s, f, g) \leq \mu_f \tag{3}$$

for $s \leq \mu_f$, and symmetrically for $s \geq \mu_f$. The posterior mean under the less precise noise density g is thus closer to the prior mean, for any signal realization.

As an illustration of Theorem 1, consider two agents who observe the same signal, such as an economic forecast, but differ in their assessment of the signal precision. The agent who assumes greater precision forms a posterior mean closer to the signal, for any signal realization. Section 4 shows several further applications.

The challenge to proving the result is that a more precise noise density has two opposing effects on the posterior mean. Consider $s > \mu_f$. The posterior $p(x)$ is proportional to $f(x)g(s-x)$, so a more precise noise density $\tilde{g}$ yields a posterior $\tilde{p}$ that downweights x far from the signal s on *both* sides of s (Figure 2). On $(-\infty, s]$, the more precise posterior $\tilde{p}$ (solid) likelihood-ratio dominates p (dashed), moving the posterior mean up toward s , the desired direction. On $[s, \infty)$, the domination reverses and pulls the posterior mean down. Because the two effects oppose each other, the result does not follow directly from standard monotone comparative statics.

The proof resolves this tension by using symmetry of the noise density to recast the problem in a form where monotonicity results do apply. Writing the posterior mean as in (4) below, the comparison reduces to integrating a monotone function against the absolute error term, which the

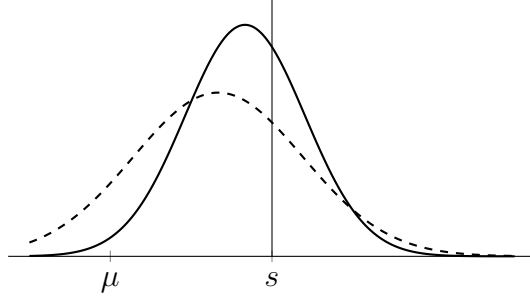


Figure 2: The posterior under a more precise noise density (solid line) and under a less precise noise density (dashed line).

precision order ranks in the likelihood-ratio order. We sketch the argument here and give the full proof in the appendix.

Let $s \geq \mu_f$, the case $s \leq \mu_f$ being analogous by symmetry. Changing variables to $\varepsilon := s - x$ and using symmetry of g around 0,

$$\begin{aligned} m(s, f, g) &= \frac{\int_{-\infty}^{\infty} (s - \varepsilon) f(s - \varepsilon) g(\varepsilon) d\varepsilon}{\int_{-\infty}^{\infty} f(s - \varepsilon) g(\varepsilon) d\varepsilon} = s - \frac{\int_0^{\infty} \varepsilon (f(s - \varepsilon) - f(s + \varepsilon)) g(\varepsilon) d\varepsilon}{\int_0^{\infty} (f(s - \varepsilon) + f(s + \varepsilon)) g(\varepsilon) d\varepsilon} \\ &= s - \int_0^{\infty} h(\varepsilon, s) \nu_g(\varepsilon) d\varepsilon, \end{aligned} \quad (4)$$

where

$$h(\varepsilon, s) := \varepsilon \frac{f(s - \varepsilon) - f(s + \varepsilon)}{f(s - \varepsilon) + f(s + \varepsilon)} \quad (5)$$

and ν_g is the density on $[0, \infty)$ proportional to $(f(s - \varepsilon) + f(s + \varepsilon)) g(\varepsilon)$. The proof establishes that $h(\varepsilon, s)$ is nondecreasing in ε on $[0, \infty)$ using symmetry and log-concavity of f . If $\tilde{g}$ is more precise than g , then g likelihood-ratio dominates $\tilde{g}$ on $[0, \infty)$, and multiplying both noise densities by $f(s - \varepsilon) + f(s + \varepsilon)$ preserves this. Hence ν_g likelihood-ratio dominates $\nu_{\tilde{g}}$, and in particular first-order stochastically dominates it. Integrating the nondecreasing h then yields a larger integral under ν_g than under $\nu_{\tilde{g}}$, which by (4) gives $m(s, f, g) \leq m(s, f, \tilde{g})$.

For the other direction, suppose $\tilde{g}$ is not more precise than g , so $\tilde{g}(x)/g(x)$ is increasing at some $x > 0$. Under a prior f concentrated around 0 and the signal realization $s = x > 0 = \mu_f$, the posterior under g then likelihood-ratio dominates the posterior under $\tilde{g}$, reversing the ordering (2). Symmetry of g and $\tilde{g}$ around 0 follows separately, by applying (2) at $s = \mu_f = 0$ as in the proof of Proposition 1. The construction uses only uniform priors, so (2) already fails for any prior class containing the uniform densities when g and $\tilde{g}$ are not both symmetric around 0 and ranked by the precision order.

The two assumptions on the prior, symmetry and log-concavity, enter the proof of Theorem 1 only through properties of $h(\varepsilon, s)$. For $s \geq \mu_f$, whenever $h(\cdot, s)$ is nonnegative and nondecreasing on the noise support, the argument delivers (2) at that signal, and symmetrically for $s \leq \mu_f$. These properties depend on the prior and the signal but not on the noise densities. The ordering (2) can hold at some signals with neither symmetry nor log-concavity of the prior. However, symmetry

is necessary if we require the ordering at every signal, and log-concavity cannot be weakened to quasi-concavity. Appendix 6.3 develops these points.

Theorem 1 shows that the familiar ordering (2) holds far more generally than for the normal distribution. No functional-form assumption is needed, only common distributional ones. The effect is thus robust across functional forms, but symmetry cannot be relaxed, which is a strong assumption. An asymmetric prior or noise density may still yield (2) at some s , especially when the asymmetry is mild, but not in general. The symmetry requirement is not specific to our comparative statics. Already the property that the posterior mean lies robustly between the prior mean and the signal realization requires symmetry, as Proposition 1 shows.

Next, we show that *scaling up* a symmetric, log-concave density results in a less precise density.

3.1 Location-Scale Experiments

Scaling up the error term by σ leads to density $g_\sigma := g(\cdot/\sigma)/\sigma$. We show comparative statics on the scale parameter $\sigma > 0$.

Lemma 1. *Let g be symmetric around 0. Then $g_{\tilde{\sigma}}$ is more precise than g_σ for all $\sigma > \tilde{\sigma} > 0$, if and only if $g(e^x)$ is log-concave. In particular, $g(e^x)$ is log-concave if $g(x)$ is log-concave.*

The condition in Lemma 1 is easy to verify and holds broadly. If g is symmetric around 0 and log-concave, then $g(e^x)$ is log-concave. Thus, all location-scale experiments with symmetric, log-concave error densities are covered, including the normal, logistic, double-exponential, and uniform.³ The condition also holds beyond log-concavity: in Appendix 6.4, we verify it for the Student- t and double-Pareto distribution.

Together, Theorem 1, Lemma 1 and Proposition 1 imply the following result. By Lemma 1, the result still holds if we weaken the assumption that the noise density g is log-concave to $\log g(e^x)$ being concave.

Corollary 1. *Let f be symmetric and log-concave and g symmetric around 0 and log-concave. Then, for all $\tilde{\sigma}, \sigma \in \mathbb{R}$ with $\sigma > \tilde{\sigma} > 0$,*

$$\begin{aligned} \forall s \leq \mu_f: \quad s &\leq m(s, f, g_{\tilde{\sigma}}) \leq m(s, f, g_\sigma) \leq \mu_f \\ \forall s \geq \mu_f: \quad s &\geq m(s, f, g_{\tilde{\sigma}}) \geq m(s, f, g_\sigma) \geq \mu_f \end{aligned} \tag{6}$$

The corollary states that the posterior mean is weakly closer to the signal realization under a smaller scale parameter.

3.2 Comparison to other Information Orders

The precision order neither implies nor is implied by the Blackwell order. Blackwell domination is very restrictive for location experiments: by Boll (1955), one location experiment Blackwell-

³ The uniform distribution has bounded support and so violates Assumption 1. However, any uniform distribution can be approximated by a log-concave distribution with full support (see the proof of Lemma 2), so our results extend to uniform errors.

dominates another if and only if the latter's noise term is obtained from the former's by adding independent noise. For $c > 1$, this makes $U[-1, 1]$ Blackwell-dominate $U[-c, c]$ only when c is an integer. The precision order instead ranks $U[-1, 1]$ as more precise than $U[-c, c]$ for every $c > 1$, as it does any scaling of a symmetric log-concave density. Hence the precision order does not imply the Blackwell order.

Perhaps surprisingly, the Blackwell order does not imply the precision order. For a counterexample, let $\tilde{\varepsilon} = \varepsilon + \varepsilon'$ be the independent sum of a standard Cauchy ε and $\varepsilon' \sim U[-1, 1]$. The experiment with noise ε Blackwell-dominates the one with noise $\tilde{\varepsilon}$, yet the density of $\tilde{\varepsilon}$ is not less precise than the one of ε . By Theorem 1, Blackwell domination is therefore not sufficient for the attenuation ordering (2).

The precision order strengthens the Lehmann order on log-concave location experiments, which ranks experiments by their value in all *monotone* decision problems (Lehmann 1988; Quah & Strulovici 2009). This order is defined for location experiments only under log-concave noise, the case satisfying the monotone likelihood-ratio property. Such an experiment dominates another in the Lehmann order if its noise is smaller in the dispersive order, meaning any two quantiles are weakly closer: CDF $\tilde{G}$ is smaller than CDF G in the dispersive order if

$$\forall 0 < \alpha < \beta < 1 : \tilde{G}^{-1}(\beta) - \tilde{G}^{-1}(\alpha) \leq G^{-1}(\beta) - G^{-1}(\alpha),$$

where G^{-1} and $\tilde{G}^{-1}$ are the right-continuous inverses of G and $\tilde{G}$. The precision order implies this. Truncated to positive values the precision order gives the likelihood-ratio order, which implies the hazard-rate order (Shaked & Shanthikumar 2007, Theorem 1.C.1), which under log-concavity implies the dispersive order (Bagai & Kochar 1986, Theorem 1). Symmetry extends the inequality from $1/2 \leq \alpha < \beta < 1$ to all $0 < \alpha < \beta < 1$.

4 Applications

We apply Theorem 1 to comparative statics for the median posterior mean, prior precision, the value of information, and the average posterior mean under overconfidence.

4.1 Median Posterior Mean

In many settings the researcher observes the state x and the population distribution of posterior means, but not the individual signals. For example, the researcher observes realized inflation together with the forecasts made a period earlier. The median of such a population admits a clean characterization under symmetric log-concave noise. We treat the average posterior mean in the Online Appendix.

Let $\hat{m}(x, f, g)$ denote the median posterior mean under state x , the median of $m(s, f, g)$ under the signal density $g(\cdot - x)$. Taking the agents' signals to be independent conditional on the state, the empirical median of their posterior means converges to $\hat{m}(x, f, g)$ as the population grows large. When g is symmetric around 0, the median signal given state x equals x . Under a strictly log-concave

noise density g the location experiment satisfies the strict monotone likelihood-ratio property, so $m(s, f, g)$ is strictly increasing in s and $\hat{m}(x, f, g) = m(x, f, g)$. This identity reduces the next result to Theorem 1 at $s = x$.

Corollary 2. *Let the noise densities g and $\tilde{g}$ be symmetric around 0 and strictly log-concave. If $\tilde{g}$ is more precise than g , then for every symmetric and log-concave prior density f :*

$$\begin{aligned} \forall x \leq \mu_f: \quad & x \leq \hat{m}(x, f, \tilde{g}) \leq \hat{m}(x, f, g) \leq \mu_f \\ \forall x \geq \mu_f: \quad & x \geq \hat{m}(x, f, \tilde{g}) \geq \hat{m}(x, f, g) \geq \mu_f. \end{aligned} \tag{7}$$

In the forecasting example, a more precise signal noise density brings the median forecast closer to realized inflation, for every symmetric log-concave prior density held by forecasters. The result requires no functional-form assumption and therefore provides a better theoretical foundation for interpreting attenuated responses as a consequence of imprecise information.

The median posterior mean complements the average posterior mean of Section 7.2 and is more robust to the mapping from beliefs to actions. When the action is an increasing, possibly nonlinear, function φ of the posterior mean, the median action equals $\varphi(\hat{m}(x, f, g))$, so Corollary 2 carries over to the median action. An analogous result on the average posterior mean would carry over to the average action only for affine φ , since averaging commutes with φ only then. Moreover, the median action remains well-defined under a purely ordinal action space.⁴

Our median characterization speaks to common-interest elections under sincere voting. A large electorate chooses between two fixed alternatives $L < R$ in $\mathbb{R}$ by majority rule under common spatial preferences $u(a, x) = -(a - x)^2$ over a state $x \in \mathbb{R}$. Each voter observes a private signal $s_i = x + \varepsilon_i$ with symmetric, strictly log-concave noise density g and holds the symmetric, log-concave prior density f . A sincere voter elects $a \in \{L, R\}$ to maximize the expectation of $u(a, x)$ given her posterior over x , without conditioning on the event that her vote is pivotal.⁵ Thus, she votes for R whenever $m(s_i, f, g) > c$ with $c := (L + R)/2$. As the electorate grows, the majority elects R if and only if $\hat{m}(x, f, g) > c$, assuming ties are broken toward L . Attenuation of $\hat{m}(x, f, g)$ toward μ_f biases the election toward the alternative closer to the prior mean, so the election fails to aggregate information efficiently. Suppose $\mu_f < c$ and that a threshold $x^* > c$ with $\hat{m}(x^*, f, g) = c$ exists. On every state $x \in (c, x^*)$ the alternative closer to the state is R , yet the majority elects L because $\hat{m}(x, f, g) < c$, so elections over-select the alternative closer to the prior mean. By Corollary 2 a more precise noise density shrinks (c, x^*) and thereby reduces the set of states on which the electorate errs. Interestingly, perceived precision acts the same way: an electorate that treats the noise density as more precise than the objective $\hat{g}$ shifts the pivotal report $\hat{m}(x, f, g)$ toward x and more often elects the alternative closer to the state.

⁴Enke, Graeber, Oprea, and Yang (2026) work with the average action, defined as the posterior mean of the optimal policy (Ilut & Valchev 2023).

⁵Under strategic voting, comparative statics in precision are uninteresting: an equilibrium exists in which information is aggregated efficiently and the alternative closer to x wins with probability approaching one as the electorate grows, irrespective of precision (McLennan 1998). Esponda and Vespa (2014) find that a majority of subjects vote non-strategically in common-interest voting experiment.

4.2 Prior Precision and Value of Information

Location experiments exhibit a useful duality between the prior and the noise density, captured by

$$m(s, f, g) = s - m(s, g, f). \quad (8)$$

Applying Theorem 1 with the roles of prior and noise interchanged yields a dual result for making the prior more precise.

Corollary 3. *For any $f, \tilde{f}$, and $\mu \in \mathbb{R}$, the following are equivalent:*

1. *Both f and $\tilde{f}$ are symmetric around μ , and $\tilde{f}$ is more precise than f .*
2. *For every symmetric around 0 and log-concave g :*

$$\begin{aligned} \forall s \leq \mu: \quad m(s, f, g) &\leq m(s, \tilde{f}, g) \leq \mu \\ \forall s \geq \mu: \quad m(s, f, g) &\geq m(s, \tilde{f}, g) \geq \mu. \end{aligned} \quad (9)$$

Corollary 3 describes how prior precision moves posterior means. We now turn to a consequence for the value of information. Classic orders such as the Blackwell and Lehmann orders compare signal structures for a fixed prior. We instead fix the signal structure and vary the prior, and show that in mean-measurable decision problems a more precise prior lowers the value of every location experiment with symmetric, log-concave noise.

The agent chooses an action $a \in \mathcal{A} \subseteq \mathbb{R}$ from a compact set, with utility

$$u(a, x) = \alpha(a) + \beta(a)x + \gamma(x),$$

where $\alpha, \beta, \gamma: \mathbb{R} \rightarrow \mathbb{R}$ are continuous and $\int |\gamma(x)|f(x)dx < \infty$. The agent's prior over the state x has density f with mean μ .

For example, estimating x under quadratic loss $u(a, x) = -(a - x)^2$ corresponds to $\alpha(a) = -a^2$, $\beta(a) = 2a$, $\gamma(x) = -x^2$. Deciding whether to buy a good of quality x at price p corresponds to $\mathcal{A} = \{0, 1\}$ and $u(a, x) = (x - p)a$, that is $\alpha(a) = -pa$, $\beta(a) = a$, $\gamma(x) = 0$.

The value of a location experiment with noise density g under prior f is the expected utility attainable with the location experiment minus the expected utility attainable without it,

$$v(f, g) := \max_{\hat{a}: \mathbb{R} \rightarrow \mathcal{A}} \int \int u(\hat{a}(s), x) g(s - x) f(x) ds dx - \max_{a \in \mathcal{A}} \int u(a, x) f(x) dx.$$

For these decision problems, the value of an experiment depends on the experiment only through the induced distribution of posterior means. Representing a signal structure by its distribution $\tau \in \Delta(\Delta(\mathbb{R}))$ over posteriors $\pi \in \Delta(\mathbb{R})$, the value is

$$\begin{aligned} &\mathbb{E}_\tau \left[\max_{a \in \mathcal{A}} \mathbb{E}_\pi [u(a, x)] \right] - \max_{a \in \mathcal{A}} \int u(a, x) f(x) dx \\ &= \mathbb{E}_\tau \left[\max_{a \in \mathcal{A}} (\alpha(a) + \beta(a) \mathbb{E}_\pi [x]) \right] - \max_{a \in \mathcal{A}} (\alpha(a) + \beta(a) \mu), \end{aligned} \quad (10)$$

since the action-independent term cancels by Bayes plausibility, $\mathbb{E}_\tau[\mathbb{E}_\pi[\gamma(x)]] = \int \gamma(x)f(x) dx$.

A more precise location experiment raises the value of information among log-concave location experiments. This is because the precision order strengthens the Lehmann order on such experiments (Lehmann 1988; Quah & Strulovici 2009), and ordering actions by $\beta(a)$ makes each mean-measurable problem a monotone decision problem. The next result shows that making the *prior* more precise reduces the value of location experiments under symmetric distributions.

Proposition 2. *Let f and $\tilde{f}$ be symmetric around μ , and let g be symmetric and log-concave. If $\tilde{f}$ is more precise than f , then $v(\tilde{f}, g) \leq v(f, g)$.*

The value of an experiment increases in how far the induced posterior means spread from μ . A more precise prior compresses the spread of posterior means on two fronts. The posterior mean reacts less to each signal realization (Corollary 3), and the prior places less weight on extreme states, so signals are less dispersed.

5 Conclusion

We extended the attenuation effect of imprecise information beyond the normal-normal model. Introducing a new precision order based on likelihood-ratio dominance, we showed that greater imprecision attenuates the posterior mean toward the prior mean across a broad class of distributions common in economic modeling. The order is not merely sufficient for this attenuation ordering but also necessary, so the precision order is characterized by the effect. The characterization provides a more robust foundation for interpreting attenuated behavior as a consequence of imprecise information, whether observational or cognitive, and pins down symmetry and log-concavity as the essential distributional conditions.

Our results also speak to signal-extraction problems more broadly. The median posterior mean inherits the same attenuation ordering under strictly log-concave noise. Applied to common-interest elections under sincere voting, attenuation of the median posterior mean biases majority choice toward the alternative nearer the prior mean, and a more precise signal shrinks the set of states on which the majority errs. Making the prior more precise, rather than the signal, moves the posterior mean in the opposite direction, toward the prior mean, and lowers the value of information.

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6 Appendix: Proofs

Lemma 2. *A uniform prior density f on an interval $[a, b]$ can be approximated by a sequence of symmetric, log-concave densities $(f_n)_{n \in \mathbb{N}}$ satisfying Assumption 1 such that $m(s, f_n, g) \rightarrow m(s, f, g)$ for any s and g satisfying Assumption 1. If f is additionally symmetric around 0, then all f_n can be taken symmetric around 0. An analogous statement holds for uniform error densities g on $[a, b]$.*

Proof. Define the prior density f_n by

$$\log f_n(x) = \begin{cases} c, & x \in [a, b] \\ c - n(x - b)^2, & x > b \\ c - n(x - a)^2, & x < a, \end{cases} \quad (11)$$

where c normalizes f_n to integrate to 1. Each f_n is symmetric, log-concave, satisfies Assumption 1, and converges pointwise to the uniform density f on $[a, b]$ as $n \rightarrow \infty$. If f is symmetric around 0, then each f_n is symmetric around 0. We claim dominated convergence gives

$$m(s, f_n, g) = \frac{\int x f_n(x) g(s - x) dx}{\int f_n(x) g(s - x) dx} \xrightarrow{n \rightarrow \infty} \frac{\int x f(x) g(s - x) dx}{\int f(x) g(s - x) dx} = m(s, f, g).$$

Both integrands below are dominated by absolutely integrable functions using $f_n(x) \leq 1/(b - a)$ and that g satisfies Assumption 1. Finally, the denominator limit is strictly positive, so the ratio converges. The argument for uniform noise on $[a, b]$ is identical. $\square$

6.1 Proposition 1

Proof. We prove the three statements of Proposition 1 in order.

Chambers and Healy (2012, Proposition 3) show that if f is symmetric and quasi-concave and g is symmetric around 0 and quasi-concave, then the posterior mean $m(s, f, g)$ lies weakly between the signal realization s and the prior mean μ_f .

Chambers and Healy (2012, Proposition 7) show that if (1) holds for all uniform g symmetric around 0, then f is symmetric and quasi-concave. By Lemma 2, uniform g is a pointwise limit of symmetric, log-concave g_n satisfying Assumption 1 with $m(s, f, g_n) \rightarrow m(s, f, g)$, so (1) extends to uniform g and the second statement follows.

Suppose that (1) holds for all symmetric, quasi-concave f . Subtracting μ_f from each side of (1) and defining $f'(\cdot) := f(\cdot + \mu_f)$ and $s' := s - \mu_f$, the hypothesis becomes: for all symmetric around 0, quasi-concave f' ,

$$\begin{aligned} \forall s' \leq 0: \quad & s' \leq m(s', f', g) \leq 0, \\ \forall s' \geq 0: \quad & s' \geq m(s', f', g) \geq 0. \end{aligned} \quad (12)$$

Let $f' \sim U[-\sigma, \sigma]$. By Lemma 2, (12) extends to this f' . Then

$$\forall \sigma > 0: \quad 0 = m(0, f', g) = \frac{\int_{-\sigma}^{\sigma} x g(-x) dx}{\int_{-\sigma}^{\sigma} g(-x) dx} = \frac{\int_{-\sigma}^{\sigma} (-x) g(x) dx}{\int_{-\sigma}^{\sigma} g(x) dx},$$

where the last equality substitutes $x \mapsto -x$. The right-hand side converges to $-\mu_g$ as $\sigma \rightarrow \infty$, so $\mu_g = 0$. By the duality $m(s', f', g) = s' - m(s', g, f')$ (see (8)),

$$\begin{aligned} s' \leq m(s', f', g) &\Leftrightarrow m(s', g, f') \leq 0 = \mu_g, \\ m(s', f', g) \leq 0 &\Leftrightarrow s' \leq m(s', g, f'). \end{aligned}$$

Combining with (12), for all symmetric around 0, quasi-concave f' ,

$$\begin{aligned} \forall s' \leq \mu_g: \quad s' &\leq m(s', g, f') \leq \mu_g, \\ \forall s' \geq \mu_g: \quad s' &\geq m(s', g, f') \geq \mu_g. \end{aligned}$$

By the second statement of the proposition (viewing g as the prior and f' as the noise density), g is symmetric and quasi-concave. Since $\mu_g = 0$, g is symmetric around 0. $\square$

6.2 Theorem 1

Proof. 1 $\Rightarrow$ 2: Let $s \geq \mu$, the case $s \leq \mu$ being analogous by symmetry. Recall from (4) the representation $m(s, f, g) = s - \int_0^\infty h(\varepsilon, s) \nu_g(\varepsilon) d\varepsilon$, where ν_g is the density on $[0, \infty)$ proportional to $(f(s + \varepsilon) + f(s - \varepsilon))g(\varepsilon)$ and

$$h(\varepsilon, s) := \varepsilon \frac{f(s - \varepsilon) - f(s + \varepsilon)}{f(s - \varepsilon) + f(s + \varepsilon)} = \varepsilon \frac{1 - f(s + \varepsilon)/f(s - \varepsilon)}{1 + f(s + \varepsilon)/f(s - \varepsilon)}. \quad (13)$$

We claim $h(\varepsilon, s)$ is nonnegative and nondecreasing in ε on $[0, \infty)$.

Nonnegativity: By symmetry of f around μ and quasi-concavity (which is implied by log-concavity and positivity), $f(x)$ is greater the closer x is to μ . For $s \geq \mu$ and $\varepsilon \geq 0$, the point $s - \varepsilon$ is weakly closer to μ than is $s + \varepsilon$, so $f(s - \varepsilon) \geq f(s + \varepsilon)$. Hence $h(\varepsilon, s) \geq 0$.

Monotonicity: By the second expression in (13), it suffices to show that the ratio $f(s + \varepsilon)/f(s - \varepsilon)$ is nonincreasing in ε on $[0, \infty)$, or equivalently that its logarithm is. By symmetry of f around μ , $f(s - \varepsilon) = f(\mu + (\mu - (s - \varepsilon))) = f(2\mu - s + \varepsilon)$, so

$$\log \frac{f(s + \varepsilon)}{f(s - \varepsilon)} = \log f(s + \varepsilon) - \log f(2\mu - s + \varepsilon).$$

Then

$$\frac{d}{d\varepsilon} \log \frac{f(s + \varepsilon)}{f(s - \varepsilon)} = (\log f)'(s + \varepsilon) - (\log f)'(2\mu - s + \varepsilon) \leq 0,$$

where the inequality uses concavity of $\log f$ together with $s + \varepsilon \geq 2\mu - s + \varepsilon$, which holds since $s \geq \mu$.

Now compare g with the more precise $\tilde{g}$. The precision order requires $\tilde{g}(\varepsilon)/g(\varepsilon)$ to be nonincreas-

ing in ε on $[0, \infty)$, that is, g likelihood-ratio dominates $\tilde{g}$ on $[0, \infty)$. Multiplying both densities by the common factor $f(s + \varepsilon) + f(s - \varepsilon)$ preserves likelihood-ratio domination, so ν_g likelihood-ratio dominates $\nu_{\tilde{g}}$, and in particular first-order stochastically dominates it. Since h is nondecreasing,

$$\int_0^\infty h(\varepsilon, s) \nu_g(\varepsilon) d\varepsilon \geq \int_0^\infty h(\varepsilon, s) \nu_{\tilde{g}}(\varepsilon) d\varepsilon,$$

and combining with (4) gives $m(s, f, g) \leq m(s, f, \tilde{g})$.

2 $\Rightarrow$ 1: We first show that if (2) holds for every symmetric and log-concave f , then $\tilde{g}$ and g are symmetric around 0. This follows by the same argument as in the proof of Proposition 6 of Chambers and Healy (2012), with the roles of the prior and noise densities swapped: applied to the uniform prior f^U and the signal $s = \mu_{f^U} = 0$, the ordering (2) forces $m(0, f^U, g) = 0$, which after differentiating in δ gives $g(-\delta) = g(\delta)$ for all $\delta > 0$. The approximation of Lemma 2 extends this to symmetric, log-concave, strictly positive priors. The argument is analogous for $\tilde{g}$.

Next we show that $\tilde{g}$ is more precise than g . Suppose, by contraposition, that $\tilde{g}$ is not more precise than g . Then the likelihood ratio $\tilde{g}(x)/g(x)$ has a strictly positive derivative at some $x = s > 0$, and by continuous differentiability, on some interval $[s - \delta, s + \delta]$ with $\delta > 0$.

Take the uniform prior $f^U(x) = \frac{1}{2\delta} \mathbf{1}(x \in [-\delta, \delta])$. Under this prior, the posterior densities upon observing s are proportional to $g(s - x)$ and $\tilde{g}(s - x)$ on $[-\delta, \delta]$ and 0 elsewhere. The ratio of these posterior densities,

$$\frac{\tilde{g}(s - x)}{g(s - x)},$$

is strictly decreasing in x on $[-\delta, \delta]$, since $\tilde{g}(\cdot)/g(\cdot)$ is strictly increasing on $[s - \delta, s + \delta]$. Hence the posterior under $\tilde{g}$ strictly likelihood-ratio dominates the posterior under g , and so

$$m(s, f^U, \tilde{g}) > m(s, f^U, g),$$

in violation of (2) at $s > 0 = \mu_{f^U}$.

By Lemma 2, f^U is a pointwise limit of symmetric, log-concave f_n satisfying Assumption 1 with $m(s, f_n, g) \rightarrow m(s, f^U, g)$ and $m(s, f_n, \tilde{g}) \rightarrow m(s, f^U, \tilde{g})$. For n large the strict ordering persists, so (2) fails for a symmetric, log-concave prior. $\square$

6.3 Assumptions on the Prior

Both symmetry and log-concavity of the prior enter the proof of Theorem 1 only through $h(\varepsilon, s)$ in the representation (4) of the posterior mean. For $s \geq \mu_f$, whenever $h(\cdot, s)$ is nonnegative and nondecreasing on the noise support, the argument of Theorem 1 yields (2) at s , and symmetrically for $s \leq \mu_f$. We give a condition delivering these properties at some s without either symmetry or log-concavity, then show symmetry is necessary for (2) to hold at every signal and that log-concavity cannot be weakened to quasi-concavity.

Suppose the noise density has bounded support $[-\bar{\varepsilon}, \bar{\varepsilon}]$, $s \geq \mu_f$, and f is nonincreasing on $[s - \bar{\varepsilon}, s + \bar{\varepsilon}]$, the states from which the signal may have originated. For $\varepsilon \geq 0$, $f(s - \varepsilon) \geq f(s + \varepsilon)$,

and as ε increases $f(s - \varepsilon)$ rises while $f(s + \varepsilon)$ falls, so $f(s + \varepsilon)/f(s - \varepsilon)$ is nonincreasing in ε . Hence

$$h(\varepsilon, s) = \varepsilon \frac{f(s - \varepsilon) - f(s + \varepsilon)}{f(s - \varepsilon) + f(s + \varepsilon)} = \varepsilon \frac{1 - f(s + \varepsilon)/f(s - \varepsilon)}{1 + f(s + \varepsilon)/f(s - \varepsilon)}$$

is nonnegative and nondecreasing on the support of the noise, and the argument of Theorem 1 yields (2) at this s , requiring neither symmetry nor log-concavity. A leading case is a quasi-concave prior, which is nonincreasing to the right of its mode, so the condition holds for any signal far enough above the mode relative to the noise support.

Symmetry of the prior is necessary for (2) to hold at every signal. Normalize $\mu_f = 0$ and take $s = 0$. Then (2) forces $m(0, f, g) = 0$. The proof of Proposition 1 shows that $m(0, f, g) = 0$ for all symmetric g if and only if f is symmetric. Thus, symmetry of the prior cannot be dropped, even though the positive example above shows that (2) can still hold at signals far from the prior mean.

The following counterexample shows that log-concavity cannot be weakened to quasi-concavity. Under a symmetric, quasi-concave but non-log-concave prior, a more precise noise density can yield a posterior mean strictly farther from the signal than a less precise one, violating (2).

Consider the prior density (up to normalization)

$$f(x) \propto \begin{cases} 2 - |x + 1| & \text{if } |x + 1| \leq 1, \\ 1 & \text{if } 1 \leq |x + 1| \leq 10, \\ 0 & \text{otherwise.} \end{cases}$$

The density is symmetric and quasi-concave around the prior mean $\mu_f = -1$, but the kinks at $|x + 1| = 1$ prevent log-concavity: the density has a strict peak around its mean on top of a flat plateau. Compare, at the signal realization $s = 0$, the noise density $\tilde{g}$ uniform on $[-1, 1]$ with the less precise g uniform on $[-5, 5]$. Direct computation gives

$$m(0, f, \tilde{g}) = -\frac{2}{15} < m(0, f, g) = -\frac{1}{11}.$$

Since $\mu_f = -1 < 0 = s$, the less precise g yields a posterior mean strictly closer to the signal than the more precise $\tilde{g}$, contradicting (2).

6.4 Lemma 1

Proof. For any $\sigma > 0$, by symmetry of g_σ around 0, truncating g_σ to $[0, \infty)$ gives the density $2g_\sigma$. By definition, $g_{\tilde{\sigma}}$ is more precise than g_σ if and only if on $[0, \infty)$ the density $2g_{\tilde{\sigma}}$ is smaller in the likelihood-ratio order than $2g_\sigma$. Rescaling by $1/\sigma$ turns this into the comparison between densities $2g_{\tilde{\sigma}/\sigma}$ and $2g$ on $[0, \infty)$. As $\sigma > \tilde{\sigma} > 0$ ranges freely, $\tilde{\sigma}/\sigma$ ranges over all of $(0, 1)$. For a nonnegative random variable with density h , scaling it by a yields a smaller variable in the likelihood-ratio order for all $a \in (0, 1)$ if and only if $\log h(e^x)$ is concave (Hu, Nanda, Xie, & Zhu 2004, Remark 5 (iii)). Applied to $h = 2g$ this gives the first claim.

For the second claim, if g is log-concave and symmetric around 0, then

$$(\log g \circ \exp)''(x) = e^x (\log g)'(e^x) + e^{2x} (\log g)''(e^x) < 0,$$

since $(\log g)'' < 0$ by log-concavity and $(\log g)'(y) < 0$ for $y > 0$ by symmetry and log-concavity. $\square$

6.5 Corollary 3

Proof. The identity (8) follows from the change of variables $\varepsilon = s - x$:

$$\begin{aligned} m(s, f, g) &= \frac{\int x f(x) g(s - x) dx}{\int f(x) g(s - x) dx} = \frac{\int (s - \varepsilon) f(s - \varepsilon) g(\varepsilon) d\varepsilon}{\int f(s - \varepsilon) g(\varepsilon) d\varepsilon} \\ &= s - \frac{\int \varepsilon g(\varepsilon) f(s - \varepsilon) d\varepsilon}{\int g(\varepsilon) f(s - \varepsilon) d\varepsilon} = s - m(s, g, f). \end{aligned}$$

1 $\Rightarrow$ 2: Let f and $\tilde{f}$ be symmetric around μ with $\tilde{f}$ more precise than f . Then $h(\cdot) := f(\cdot + \mu)$ and $\tilde{h}(\cdot) := \tilde{f}(\cdot + \mu)$ are symmetric around 0 and $\tilde{h}$ is more precise than h . By Theorem 1 applied with h and $\tilde{h}$ as noise densities, for every symmetric around 0 and log-concave g ,

$$\begin{aligned} \forall s \leq 0: \quad s &\leq m(s, g, \tilde{h}) \leq m(s, g, h) \\ \forall s \geq 0: \quad s &\geq m(s, g, \tilde{h}) \geq m(s, g, h). \end{aligned} \tag{14}$$

By (8), this is equivalent to

$$\begin{aligned} \forall s \leq 0: \quad m(s, h, g) &\leq m(s, \tilde{h}, g) \leq 0 \\ \forall s \geq 0: \quad m(s, h, g) &\geq m(s, \tilde{h}, g) \geq 0. \end{aligned} \tag{15}$$

By $m(s, h, g) = m(s + \mu, f, g) - \mu$, this is equivalent to (9).

2 $\Rightarrow$ 1: Assume f , $\tilde{f}$ and μ are such that (9) holds for every symmetric around 0 and log-concave g . Define h and $\tilde{h}$ as above. Reversing the steps of the first direction, (9) is equivalent to (14). We now apply the (2) $\Rightarrow$ (1) direction of Theorem 1 with g as prior and $h, \tilde{h}$ as noise densities. That direction's hypothesis requires the corresponding inequalities to hold for every symmetric and log-concave prior, but inspection of its proof shows that only symmetric around 0 and log-concave priors are used, so the weaker hypothesis suffices. We conclude that h and $\tilde{h}$ are symmetric around 0 and $\tilde{h}$ is more precise than h . Translating back, f and $\tilde{f}$ are symmetric around μ and $\tilde{f}$ is more precise than f . $\square$

6.6 Proposition 2

Proof. Without loss, g is symmetric around 0: if g is symmetric around y , then $m(s, f, g) = m(s, f, \hat{g}) - y$ for $\hat{g}(\cdot) := g(\cdot - y)$, and $v(f, g) = v(f, \hat{g})$.

By (10), $v(f, g)$ depends only on the distribution of posterior means. Let $H_{f,g}$ be the CDF of the posterior mean under noise g and prior f . The signal distribution is the convolution of prior

and noise, and the posterior-mean distribution is the pushforward of the signal distribution under $s \mapsto m(s, f, g)$. Define $w: \mathbb{R} \rightarrow \mathbb{R}$ by

$$w(m) := \max_{a \in \mathcal{A}} (\alpha(a) + \beta(a) m),$$

which is convex as a maximum of affine functions. The value can be written as

$$v(f, g) = \int w(m) dH_{f,g}(m) - w(\mu).$$

Bayes plausibility gives $\int m dH_{f,g}(m) = \mu$, so adding an affine function to w leaves $v(f, g)$ unchanged. Subtracting a supporting line of w at μ , we may assume w is minimal at μ . Since f and g are symmetric around μ and 0, $H_{f,g}$ is symmetric around μ , so

$$\int w(m) dH_{f,g}(m) = \int \frac{1}{2} (w(m) + w(2\mu - m)) dH_{f,g}(m),$$

and the symmetrized integrand is convex, symmetric around μ , and minimal at μ . We may thus assume w has these three properties.

Writing out the distribution $H_{f,g}$,

$$v(f, g) = \int \underbrace{\int w(m(s, f, g)) g(s - x) ds}_{=: W_{f,g}(x)} f(x) dx - w(\mu).$$

We decompose the change of the value into two parts:

$$v(\tilde{f}, g) - v(f, g) = \underbrace{\int (W_{\tilde{f},g}(x) - W_{f,g}(x)) \tilde{f}(x) dx}_{=: A} + \underbrace{\int W_{f,g}(x) (\tilde{f}(x) - f(x)) dx}_{=: B}.$$

First, we prove $A \leq 0$. Fix a state x . The signal density $g(s - x)$ is unchanged, and by Corollary 3 the posterior mean under the more precise prior is weakly closer to μ and on the same side of μ : $\mu \leq m(s, \tilde{f}, g) \leq m(s, f, g)$ for $s \geq \mu$, symmetrically for $s \leq \mu$. Since w is convex and minimal at μ , w is nondecreasing in $|m - \mu|$, so $w(m(s, \tilde{f}, g)) \leq w(m(s, f, g))$ for every s . So $W_{\tilde{f},g} \leq W_{f,g}$ pointwise and $A \leq 0$.

Second, we prove $B \leq 0$. We first show $W_{f,g}$ is nondecreasing on $[\mu, \infty)$ and symmetric around μ . Under log-concave noise the location experiment has the monotone likelihood-ratio property, so $m(s, f, g)$ is nondecreasing in s . By symmetry of the prior and noise, $m(\cdot, f, g)$ is antisymmetric around μ , so $m(\mu, f, g) = \mu$ and hence $m(s, f, g) \geq \mu$ for $s \geq \mu$. Therefore $|m(s, f, g) - \mu|$ is nondecreasing in $|s - \mu|$. As w is convex, symmetric around μ , and minimal at μ , w is nondecreasing in $|m - \mu|$, so $w(m(s, f, g)) = \psi(|s - \mu|)$ for some nondecreasing ψ . With $s = x + \varepsilon$,

$$W_{f,g}(x) = \int w(m(s, f, g)) g(s - x) ds = \mathbb{E}[\psi(|x - \mu + \varepsilon|)], \quad \varepsilon \sim g.$$

Since ψ is nondecreasing, it suffices that $|x - \mu + \varepsilon|$ is first-order stochastically increasing in x on $[\mu, \infty)$. For $0 \leq c_1 \leq c_2$ and $t \geq 0$,

$$\Pr(|c_2 + \varepsilon| > t) - \Pr(|c_1 + \varepsilon| > t) = \int_{c_1}^{c_2} (g(t - c) - g(t + c)) dc \geq 0,$$

since g is symmetric and quasi-concave and $|t - c| \leq t + c$. Thus, $|c + \varepsilon|$ is increasing in first-order stochastic dominance in c . Hence $W_{f,g}$ is nondecreasing on $[\mu, \infty)$, and symmetric around μ by symmetry of $w(m(\cdot, f, g))$ and g .

Both f and $\tilde{f}$ are symmetric around μ , so each places mass $\frac{1}{2}$ on $[\mu, \infty)$. On $[\mu, \infty)$, $\tilde{f}$ being more precise means $\tilde{f}/f$ is nonincreasing, so the restriction of $\tilde{f}$ to $[\mu, \infty)$ is likelihood-ratio dominated by that of f , hence first-order stochastically dominated. As $W_{f,g}$ is nondecreasing on $[\mu, \infty)$,

$$\int_{\mu}^{\infty} W_{f,g}(x) \tilde{f}(x) dx \leq \int_{\mu}^{\infty} W_{f,g}(x) f(x) dx.$$

The inequality holds on $(-\infty, \mu]$ by symmetry of $W_{f,g}$ around μ , so $B \leq 0$.

Together, $v(\tilde{f}, g) - v(f, g) = A + B \leq 0$. □

7 Online Appendix

7.1 Non-log-concave location-scale experiments

The main text gives several examples of commonly encountered symmetric, log-concave distributions. For several symmetric distributions that are not log-concave, $\log g(e^x)$ is nevertheless concave.

For example, the *Student-t* distribution with parameter $\nu > 0$, which includes as a special case the Cauchy distribution, gives

$$\begin{aligned} g(x) &\propto \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} \\ \Rightarrow \log g(e^x) &= C - \frac{\nu+1}{2} \log\left(1 + \frac{1}{\nu}e^{2x}\right) \\ \Rightarrow \frac{d^2}{dx^2} \log g(e^x) &= \frac{d}{dx} - \frac{\nu+1}{2} \frac{\frac{2}{\nu}e^{2x}}{1 + \frac{1}{\nu}e^{2x}} = -\frac{\nu+1}{2} \frac{\frac{4}{\nu}e^{2x}}{(1 + \frac{1}{\nu}e^{2x})^2} < 0, \end{aligned}$$

and hence has log-concave $g(e^x)$.

Creating a symmetric distribution from the *Pareto* distribution, analogous to the double-exponential distribution, with $\alpha > 0$ gives

$$g(x) \propto x^{-\alpha-1} \Rightarrow \log g(e^x) = C - (\alpha+1)x,$$

with log-linear and hence log-concave $g(e^x)$.

7.2 Average Posterior Means

Section 4.1 characterized comparative statics of the median of the population of posterior means. We now turn to the average. Taking the agents' signals to be independent conditional on the state, the empirical average of their posterior means converges to the average posterior mean conditional on the state,

$$\bar{m}(x, f, g) := \int m(s, f, g) g(s - x) ds,$$

as the population grows large. The average is the relevant statistic when only an aggregate is observed, such as total or average consumption or investment.

We first show that the average posterior mean lies between the state and the prior mean, extending Proposition 1.

Proposition 3. *Let f be symmetric around μ and quasi-concave and g be symmetric around 0 and quasi-concave. Then*

$$\begin{aligned} \forall x \leq \mu: \quad & x \leq \bar{m}(x, f, g) \leq \mu \\ \forall x \geq \mu: \quad & x \geq \bar{m}(x, f, g) \geq \mu. \end{aligned}$$

The proof pairs each signal with its reflection across μ . By symmetry of the prior and noise

densities, $m(\cdot, f, g)$ is antisymmetric around μ , and Proposition 1 places each posterior mean between the signal and μ . Quasi-concavity of the noise makes the signal on the same side of μ as the state more likely than its reflection, which places the average between the state and the prior mean.

Overconfidence We now compare agents who misperceive the signal precision. Signals are drawn from the objective signal structure with prior density f and noise density $\hat{g}$, but an agent who perceives the noise density as g reports the posterior mean $m(s, f, g)$ on observing s . Overconfidence in the signal, also called overprecision, is pervasive (Moore and Healy 2008). Generalizing the normal-normal definition of Ortleva and Snowberg (2015), we define relative (over)confidence through the precision order: the agent perceiving g is more confident than the agent perceiving $\tilde{g}$ if g is more precise than $\tilde{g}$.

When the objective noise density is $\hat{g}$ and the agent perceives it as g , the average posterior mean conditional on state x is

$$\overline{m}(x, f, g, \hat{g}) := \int m(s, f, g) \hat{g}(s - x) ds.$$

Proposition 4. *Let f be symmetric around μ and log-concave. Let $\hat{g}, g, \tilde{g}$ be symmetric around 0 and quasi-concave. If g is more precise than $\tilde{g}$, then*

$$\begin{aligned} \forall x \leq \mu: \quad & x \leq \overline{m}(x, f, g, \hat{g}) \leq \overline{m}(x, f, \tilde{g}, \hat{g}) \leq \mu \\ \forall x \geq \mu: \quad & x \geq \overline{m}(x, f, g, \hat{g}) \geq \overline{m}(x, f, \tilde{g}, \hat{g}) \geq \mu \end{aligned} \tag{16}$$

The average posterior mean of the more confident agent is closer to the true state and farther from the prior mean. The bounds by x and μ follow as in Proposition 3. For the middle inequality, pairing each signal with its reflection expresses the difference between the two averages as an integral of two nonnegative factors: the gap between the posterior means, signed by Theorem 1, and the likelihood advantage of the signal on the state's side, signed by symmetry and quasi-concavity of $\hat{g}$.

Overprecision trades higher individual variance for lower aggregate bias. A more overprecise agent reacts more strongly to her signal, raising the variance of her posterior mean and making her individual report noisier. The same reactivity moves the average posterior mean closer to the state. Averaging over many agents washes out the added variance and retains the reduced bias, so overprecision is socially valuable when only the accuracy of the average posterior mean matters, as when firms or parties cater to average opinion.

Prior Precision Greater prior precision moves the average posterior mean opposite to greater signal confidence, toward the prior mean and away from the state.

Proposition 5. *Let f and $\tilde{f}$ be symmetric around μ and quasi-concave. For every symmetric around 0 and log-concave g , if $\tilde{f}$ is more precise than f , then*

$$\begin{aligned} \forall x \leq \mu: \quad & x \leq \overline{m}(x, f, g) \leq \overline{m}(x, \tilde{f}, g) \leq \mu \\ \forall x \geq \mu: \quad & x \geq \overline{m}(x, f, g) \geq \overline{m}(x, \tilde{f}, g) \geq \mu \end{aligned} \tag{17}$$

The bounds by x and μ follow from Proposition 3, applied to f and $\tilde{f}$ separately. For the middle inequality, pairing each signal with its reflection expresses the difference between the two averages as an integral of two nonnegative factors: the gap between the individual posterior means, signed by Corollary 3, and the likelihood advantage of the signal on the state's side, signed by symmetry and quasi-concavity of g .

7.3 Proofs

Proof of Proposition 3. For $a \in \mathbb{R}$, let $f_a := f(\cdot - a)$ denote the prior shifted by a , which has mean $\mu_{f_a} = \mu_f + a$. It is easy to verify that $m(s+a, f_a, g) = m(s, f, g) + a$, and $\bar{m}(x+a, f_a, g) = \bar{m}(x, f, g) + a$. Shifting the prior and the signal by the same amount thus shifts the average posterior mean by that amount, so it is without loss to assume $\mu_f = 0$. Let $x \geq 0$. The case $x \leq 0$ is analogous by symmetry of f and g .

Splitting the integral defining $\bar{m}$ at $s = 0$,

$$\begin{aligned}\bar{m}(x, f, g) &= \int_0^\infty (m(s, f, g) g(s-x) + m(-s, f, g) g(-s-x)) ds \\ &= \int_0^\infty m(s, f, g) (g(s-x) - g(-s-x)) ds,\end{aligned}\tag{18}$$

where we used $m(-s, f, g) = -m(s, f, g)$ due to symmetry of f and g . For $s, x \geq 0$, symmetry and quasi-concavity of g give $g(-s-x) = g(s+x) \leq g(s-x)$, and Proposition 1 gives $0 \leq m(s, f, g) \leq s$. The integrand is therefore nonnegative, proving $\bar{m}(x, f, g) \geq 0 = \mu_f$.

For the upper bound, symmetry of g around 0 gives

$$x = \int_{-\infty}^\infty s g(s-x) ds = \int_0^\infty s (g(s-x) - g(-s-x)) ds.\tag{19}$$

Comparing (18) and (19) term by term, $m(s, f, g) \leq s$ together with $g(s-x) - g(-s-x) \geq 0$ shows the integrand of (18) is at most that of (19). Hence $\bar{m}(x, f, g) \leq x$. $\square$

Proof of Proposition 4. As in the proof of Proposition 3, shifting the prior and the signal by the same amount shifts $\bar{m}$ by that amount, so it is without loss to suppose $\mu_f = 0$. Let $x \geq 0$. The case $x \leq 0$ is analogous by symmetry of f , g , $\tilde{g}$, and $\hat{g}$.

The bounds $x \geq \bar{m}(x, f, g, \hat{g})$ and $\bar{m}(x, f, \tilde{g}, \hat{g}) \geq \mu_f$ follow exactly as in Proposition 3, with the objective noise density $\hat{g}$ as the weighting density. It remains to show the middle inequality $\bar{m}(x, f, g, \hat{g}) \geq \bar{m}(x, f, \tilde{g}, \hat{g})$. Splitting each defining integral at $s = 0$ and substituting $s \mapsto -s$ on the negative part,

$$\begin{aligned}\bar{m}(x, f, g, \hat{g}) &= \int_0^\infty (m(s, f, g) \hat{g}(s-x) + m(-s, f, g) \hat{g}(-s-x)) ds, \\ \bar{m}(x, f, \tilde{g}, \hat{g}) &= \int_0^\infty (m(s, f, \tilde{g}) \hat{g}(s-x) + m(-s, f, \tilde{g}) \hat{g}(-s-x)) ds.\end{aligned}$$

Subtracting and using antisymmetry $m(-s, f, g) = -m(s, f, g)$ and $m(-s, f, \tilde{g}) = -m(s, f, \tilde{g})$, which hold by symmetry of the prior and noise densities,

$$\bar{m}(x, f, g, \hat{g}) - \bar{m}(x, f, \tilde{g}, \hat{g}) = \int_0^\infty \underbrace{(m(s, f, g) - m(s, f, \tilde{g}))}_{\geq 0} \underbrace{(\hat{g}(s-x) - \hat{g}(-s-x))}_{\geq 0} ds \geq 0.$$

The first factor is nonnegative by Theorem 1, since g is more precise than $\tilde{g}$ and $s \geq \mu_f = 0$. The second factor is nonnegative by symmetry and quasi-concavity of $\hat{g}$, which give $\hat{g}(s-x) \geq \hat{g}(s+x) = \hat{g}(-s-x)$ for $s, x \geq 0$. $\square$

Proof of Proposition 5. As in the proof of Proposition 3, shifting both priors and the signal by the same amount shifts the average posterior means by that amount, so it is without loss to suppose $\mu = 0$. We treat $x \geq 0$, the case $x \leq 0$ being analogous by symmetry of f , $\tilde{f}$, and g .

The bounds $x \geq \bar{m}(x, f, g)$ and $\bar{m}(x, \tilde{f}, g) \geq \mu$ follow from Proposition 3, applied to the priors f and $\tilde{f}$ respectively (both symmetric around μ and quasi-concave, with g symmetric and quasi-concave). It remains to show the middle inequality $\bar{m}(x, f, g) \geq \bar{m}(x, \tilde{f}, g)$. Splitting each defining integral at $s = 0$ and substituting $s \mapsto -s$ on the negative part,

$$\begin{aligned} \bar{m}(x, f, g) &= \int_0^\infty (m(s, f, g) g(s-x) + m(-s, f, g) g(-s-x)) ds, \\ \bar{m}(x, \tilde{f}, g) &= \int_0^\infty (m(s, \tilde{f}, g) g(s-x) + m(-s, \tilde{f}, g) g(-s-x)) ds. \end{aligned}$$

Subtracting and using antisymmetry $m(-s, f, g) = -m(s, f, g)$ and $m(-s, \tilde{f}, g) = -m(s, \tilde{f}, g)$, which hold by symmetry of the priors and the noise density,

$$\bar{m}(x, f, g) - \bar{m}(x, \tilde{f}, g) = \int_0^\infty \underbrace{(m(s, f, g) - m(s, \tilde{f}, g))}_{\geq 0} \underbrace{(g(s-x) - g(-s-x))}_{\geq 0} ds \geq 0.$$

The first factor is nonnegative by Corollary 3: since $\tilde{f}$ is more precise than f and g is log-concave, $m(s, f, g) \geq m(s, \tilde{f}, g)$ for $s \geq \mu = 0$. The second factor is nonnegative by symmetry and quasi-concavity of g , which give $g(s-x) \geq g(s+x) = g(-s-x)$ for $s, x \geq 0$. $\square$